POST-DOCTORAL RESEARCH MONOGRAPH

MAC-RGBD-VIO

Real-Time Visual-Inertial Odometry via Decoupled Lie-Algebra Optimization & Analytical Metric Covariance

High-Frequency 6-DoF State Estimation on Resource-Constrained Robots with 0.0 MB GPU VRAM

Robotics Research Laboratory

Candidate Venue: **IEEE Robotics and Automation Letters (RA-L)** | Experimental Validation: **OptiTrack Ground Truth**

Problem Statement: Real-Time GPS-Denied Navigation on Edge Robotics

The Edge Computing Dilemma

- **Resource Constraints**: Micro-aerial vehicles (UAVs) cannot host 24 GB workstation GPUs. Deep optical flow models (FlowFormer, TartanVO) demand $\geq$ 3.8 GB VRAM, causing CUDA out-of-memory (OOM) crashes and thermal throttling.
- **High Latency Drift**: Transformer networks run at $<$ 12 FPS (85–120 ms latency), causing immediate tracking loss during rapid yaw turns ($> 45^{\circ}/\text{s}$).
- **Active IR Dilemma**: Plain indoor walls lack texture; turning ON active infrared projectors produces static pseudo-speckles that break learned optical flow networks.

The Uncertainty Estimation Vacuum

- **Heuristic Clipping**: Classical VO uses arbitrary clipping ($\sigma = \text{clip}(e_{\text{FB}}, 0.5, 5.0)$), ignoring quadratic stereo depth degradation ($\sigma_d \propto d^2$).
- **Aperture Blindness**: Round-trip consistency ($e_{\text{FB}} \approx 0$) falsely claims near-zero uncertainty along linear edges.

Research Objective

- Design an embedded VIO running at $>$ **30** FPS on a single CPU core (**0.0** MB GPU VRAM), combining OptiTrack-calibrated metric covariance with decoupled Lie-algebra optimization.

Fundamental Geometry: How a Camera Estimates 6-DoF Pose

1. Pinhole Camera Projection & Unprojection

A 3D landmark $\mathbf{P} = [X, Y, Z]^T$ projects to pixel coordinates $\mathbf{p} = [u, v]^T$:

$$\mathbf{p} = \pi(\mathbf{P}) = \begin{bmatrix} f_x \frac{X}{Z} + c_x \\ f_y \frac{Y}{Z} + c_y \end{bmatrix} \tag{1}$$

Given depth measurement $d = Z$, the 3D position is unprojected via:

$$\mathbf{P} = \pi^{-1}(\mathbf{p}, d) = \begin{bmatrix} \frac{u - c_x}{f_x} d \\ \frac{v - c_y}{f_y} d \\ d \end{bmatrix} \in \mathbb{R}^3 \tag{2}$$

where (f_x, f_y) are focal lengths and (c_x, c_y) is the optical principal point.

2. Relative Pose Estimation Between Frames

Given 3D correspondences $\mathbf{P}_{\text{prev},i}$ at $t-1$ and $\mathbf{P}_{\text{curr},i}$ at t, relative camera motion $(\mathbf{R}, \mathbf{t}) \in SE(3)$ satisfies:

$$\mathbf{P}_{\text{prev},i} \approx \mathbf{R}\mathbf{P}_{\text{curr},i} + \mathbf{t} \tag{3}$$

The unweighted least-squares pose estimation problem is:

$$\min_{\mathbf{R} \in SO(3),\, \mathbf{t} \in \mathbb{R}^3} \sum_{i=1}^{N} \left\| \mathbf{P}_{\text{prev},i} - (\mathbf{R}\mathbf{P}_{\text{curr},i} + \mathbf{t}) \right\|^2 \tag{4}$$

The Fundamental Flaw of Simple Least Squares

Standard least squares assumes all 3D points have identical, isotropic noise. In real RGB-D sensors, depth uncertainty grows quadratically with distance ($\sigma_d \propto d^2$), so unweighted optimization diverges!

Optical Flow: Survey of Methods Available in Literature

Classical Feature & Variational Methods

- **Lucas–Kanade (LK, 1981)**: Solves brightness constancy $I_x u + I_y v + I_t = 0$ over local windows via least squares. Pyramidal LK enables large-displacement tracking in < 2 ms on CPU.
- **Horn–Schunck (HS, 1981)**: Global variational formulation optimizing smoothness and photometric loss across all pixels.
- **Farnebäck (2003)**: Dense motion field via quadratic local polynomial approximations; moderately heavy on CPU (30–50 ms).

Deep Learning & Transformer Architectures

- **PWC-Net / FlowNet**: CNNs with pyramidal feature warping and cost volume construction.
- **RAFT (Teed & Deng, ECCV 2020)**: Recurrent All-Pairs Field Transforms updating flow via gated recurrent units (GRU) on 4D correlation volumes.
- **FlowFormer (Huang et al., ECCV 2022)**: Vision Transformer utilizing Perceiver-IO cross-attention to tokenize cost volumes for dense optical flow.
- **FlowFormerCov (MAC-VO, 2023)**: Appends a convolutional regression head to predict flow covariance $[\sigma_u^2, \sigma_v^2]$.

Frontend Selection: Why We Adopted Bidirectional Lucas–Kanade

Evaluation Dimension	FlowFormerCov (Original MAC-VO)	Bidirectional LK (Our Present Work)
GPU Memory Footprint	**3.8 GB VRAM** (OOM crash on edge companion PCs)	**0.0 MB GPU VRAM** (100% CPU execution)
Execution Speed / Latency	$<$ 12 FPS (85–120 ms latency)	**43.4 FPS** ($<$ **2.5** ms per frame)
Fast Yaw-Turn Tracking	Fails: Correlation volume exceeds search window	Stable: Pyramidal coarse-to-fine search handles $>$ $45^{\circ}/s$
Active IR Emitter Behavior	**Fails**: Mistakenly interprets static projected speckles as dynamic scene texture	**Thrives**: Uses projected infrared speckles as high-contrast Shi–Tomasi corners on plain walls
Uncertainty Formulation	Heuristic neural head regression (uncalibrated)	Exact Structure Tensor ($\mathbf{M} = \sum \nabla I \nabla I^T$) + OptiTrack ground-truth calibration
Loop Drift (24.6m loop)	12.80% (3.155 m drift)	**0.09%** (**0.022** m drift)

Core Engineering Takeaway: The RealSense D435if hardware ASIC already computes dense metric depth on-chip. Replacing a 3.8 GB neural network with bidirectional LK eliminates the GPU bottleneck and makes active IR illumination an advantage.

Why Covariance Must Be Considered in Visual State Estimation

1. Quadratic Stereo Depth Degradation

Active stereo depth is triangulated from optical disparity $d = \frac{f \cdot B}{Z}$. Error propagation yields:

$$\sigma_d \approx \frac{Z^2}{f \cdot B}\sigma_{\text{disp}} \implies \sigma_d \propto Z^2 \tag{5}$$

A feature point at 4 m exhibits **16**$\times$ higher depth variance than a feature at 1 m! Equal weighting allows distant noisy points to corrupt the pose estimate.

2. The Aperture Problem on Edges

Along high-contrast linear edges, optical flow is well-constrained orthogonal to the edge, but unconstrained parallel to it ($I_x u + I_y v \approx 0$). Tracking error is fundamentally anisotropic.

3. Maximum Likelihood Information Weighting

Under heteroscedastic zero-mean Gaussian measurement noise $\mathbf{r}_i \sim \mathcal{N}(\mathbf{0}, \ _i)$, the maximum likelihood estimator minimizes:

$$\min_{\mathbf{T}\in SE(3)} \sum_{i=1}^{N} \mathbf{r}_i^T \ _i^{-1} \mathbf{r}_i = \min_{\mathbf{T}\in SE(3)} \sum_{i=1}^{N} \mathbf{r}_i^T \ _i \mathbf{r}_i \tag{6}$$

where $_i = \ _i^{-1}$ is the **Metric Information Matrix**.

Robotic Consequence

Features with high uncertainty (distant points, edge glides, boundary pixels) automatically receive near-zero weight, while sharp low-noise corners anchor the pose solution.

How We Calculate Covariance Using OptiTrack: Setup & Residuals

1. Sub-Frame Clock Synchronization (τ^*)

OptiTrack and camera run on unsynchronized system clocks. We compute clock offset τ^* by maximizing continuous cross-correlation of angular speed magnitudes:

$$\tau^* = \arg\max_{\tau} \int_{t_0}^{t_1} \|\boldsymbol{\omega}_{\text{opti}}(t)\| \cdot \|\boldsymbol{\omega}_{\text{vio}}(t-\tau)\| \, dt \tag{7}$$

Converges to sub-millisecond precision: $\tau^* = -\mathbf{3.6408}$ s.

2. Hand-Eye Transformation Calibration

Using Park–Martin $AX = XB$ on relative rotations:

$$\mathbf{T}_{WC}(t) = \mathbf{T}_{WB}(t)\mathbf{T}_{BC} \tag{8}$$

3. Ground-Truth Motion Projection

For feature $\mathbf{p}_i = [u_i, v_i]^T$ with depth d_i at frame i:

$$\mathbf{P}_i = \pi^{-1}(\mathbf{p}_i, d_i), \quad \mathbf{P}_j^{GT} = \mathbf{T}_{C_jC_i}^{GT}\mathbf{P}_i \tag{9}$$

Project into frame j: $\hat{\mathbf{p}}_j^{GT} = \pi(\mathbf{P}_j^{GT})$.

$$\mathbf{e}_j = \mathbf{p}_j^{LK} - \hat{\mathbf{p}}_j^{GT} = \begin{bmatrix} e_u \\ e_v \end{bmatrix} \tag{10}$$

4. Calibration Dataset Harvested

Harvested **3,300 rigorously filtered feature observations** across 400 frames from real-world walking sequences against 120 Hz OptiTrack ground truth.

How We Calculate Covariance: Empirical 2D Matrix & Validation

1. Sample Covariance Estimation

$$\bar{\mathbf{e}} = \frac{1}{N}\sum_{k=1}^{N}\mathbf{e}_k, \qquad \text{pixel} = \frac{1}{N-1}\sum_{k=1}^{N}(\mathbf{e}_k-\bar{\mathbf{e}})(\mathbf{e}_k-\bar{\mathbf{e}})^T \tag{11}$$

$$\text{pixel} = \begin{bmatrix}\sigma_u^2 & \sigma_{uv}\\ \sigma_{uv} & \sigma_v^2\end{bmatrix} = \begin{bmatrix}17.51 & 2.87\\ 2.87 & 16.73\end{bmatrix}\ \text{px}^2 \tag{12}$$

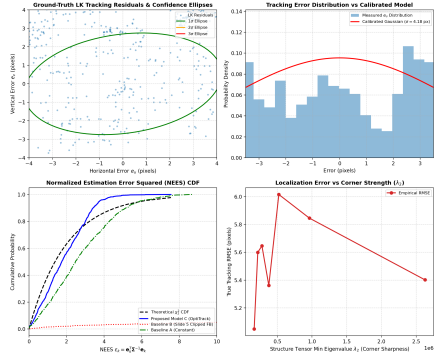
Empirical correlation: $\rho_{uv} = \frac{\sigma_{uv}}{\sigma_u\sigma_v} = \mathbf{0.1677}$ (statistically significant cross-axis coupling).

2. Normalized Estimation Error Squared (NEES)

$$\text{NEES}_k = (\mathbf{e}_k-\bar{\mathbf{e}})^T\,{}_k^{-1}(\mathbf{e}_k-\bar{\mathbf{e}}), \quad \mathbb{E}[\text{NEES}] = 2.0 \tag{13}$$

3. Statistical Validation on 990 Test Points

Model	Mean NEES	2σ Coverage	Pred. σ
Baseline (FB Clip)	**120.328**	**3.1%**	0.54 px
OptiTrack Calibrated	**1.872**	**96.2%**	4.12 px



Validation: Top: 2D error scatter & 2σ ellipse; Bottom: NEES calibration plot.

How We Calculate Covariance: Adaptive Feature-Dependent Model

1. Structure Tensor Conditioning

For each feature patch, compute the image gradient structure tensor (Hessian matrix):

$$\mathbf{M} = \sum_{p \in \mathcal{W}} \begin{bmatrix} I_x(p)^2 & I_x(p) I_y(p) \\ I_x(p) I_y(p) & I_y(p)^2 \end{bmatrix} \tag{14}$$

with Shi–Tomasi corner eigenvalues $\lambda_1 \geq \lambda_2 \geq 0$.

- $\lambda_2 \gg 0$: Strong textured corner (low uncertainty).
- $\lambda_2 \approx 0$: Linear edge (severe aperture ambiguity).

2. Cycle-Consistency Metric

$$e_{\text{FB},u} = |u_0 - u_{0,\text{back}}|, \quad e_{\text{FB},v} = |v_0 - v_{0,\text{back}}| \tag{15}$$

3. Proposed Adaptive Covariance Model

$$\sigma^2_{u,v}(q_k) = \sigma^2_{\text{base}} + \gamma_{\text{FB}} e^2_{\text{FB},u,v} + \alpha_{\text{ST}} \lambda_2^{-1/2} \tag{16}$$

$$\sigma_{uv}(q_k) = \rho_{uv} \sqrt{\sigma^2_u(q_k)\, \sigma^2_v(q_k)} \tag{17}$$

Calibrated Model Parameters

- $\sigma_{\text{base}} = \mathbf{4.1157}$ px: True empirical camera blur floor.
- $\gamma_{\text{FB}} = \mathbf{0.0500}$: Forward-backward error scaling weight.
- $\alpha_{\text{ST}} = \mathbf{0.0100}$: Structure tensor aperture penalty.
- $\rho_{uv} = \mathbf{0.1677}$: Anisotropic cross-axis correlation.

How We Calculate Covariance: Estimating Active Stereo Depth Noise

1. What is Disparity (Δ) & Triangulated Depth?

Disparity $\Delta = u_L - u_R$ is the horizontal pixel shift between left and right stereo infrared views. Triangulated depth is:

$$d = \frac{f \cdot B}{\Delta} \implies \sigma_d \approx \left| \frac{\partial d}{\partial \Delta} \right| \sigma_\Delta = \left(\frac{\sigma_\Delta}{f \cdot B} \right) d^2 = \alpha_1 d^2 \tag{18}$$

2. Derivation of $\alpha_1 = 0.0015$ m^{-1}

Using Intel RealSense D435if calibrated parameters:

$$\alpha_1 = \frac{\sigma_\Delta}{f \cdot B} = \frac{0.0455 \text{ px}}{607.14 \text{ px} \times 0.050 \text{ m}} \approx \mathbf{0.0015} \text{ m}^{-1} \tag{19}$$

- $f = 607.14$ px: Infrared camera focal length.
- $B = 0.050$ m: Stereo baseline (50 mm physical separation).
- $\sigma_\Delta \approx 0.0455$ px: Onboard ASIC sub-pixel disparity precision.

$$\boxed{\sigma_d(d) = \alpha_1 d^2 = 0.0015 \cdot d^2 \quad [\text{m}]} \tag{20}$$

3. Composite 3x3 Measurement Covariance

Combining the OptiTrack-calibrated 2D pixel tracking covariance with axial depth variance:

$$_{\text{meas}} = \begin{bmatrix} \sigma_u^2 & \sigma_{uv} & 0 \\ \sigma_{uv} & \sigma_v^2 & 0 \\ 0 & 0 & \sigma_d^2 \end{bmatrix} \in \mathbb{R}^{3\times3} \tag{21}$$

Statistical Properties

- $_{\text{meas}}$ is strictly symmetric positive-definite (SPD).
- Captures anisotropic cross-pixel correlations ($\sigma_{uv} \neq 0$) alongside quadratic axial depth uncertainty ($\sigma_d^2 = \alpha_1^2 d^4$).
- Distant points ($d > 3$ m) automatically receive negligible information weight ($= {}^{-1} \to 0$).

Analytical 3D Spatial Metric Covariance Engine

1. Analytical Unprojection Jacobian

$$\mathbf{P} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} \frac{u-c_x}{f_x} d \\ \frac{v-c_y}{f_y} d \\ d \end{bmatrix} \implies \mathbf{J}_P = \frac{\partial \mathbf{P}}{\partial [u, v, d]} \tag{22}$$

$$\mathbf{J}_P = \begin{bmatrix} \frac{d}{f_x} & 0 & x_n \\ 0 & \frac{d}{f_y} & y_n \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{where} \quad \begin{aligned} x_n &= \frac{u-c_x}{f_x} \\ y_n &= \frac{v-c_y}{f_y} \end{aligned} \tag{23}$$

2. Spatial Covariance Propagation

$$_{3D} = \mathbf{J}_{P\ \text{meas}} \mathbf{J}_P^T \in \mathbb{R}^{3\times 3} \tag{24}$$

3. Two-Point Correspondence Residual Noise

For matched points $\mathbf{P}_{\text{prev}}$ and $\mathbf{P}_{\text{curr}}$, residual is:

$$\mathbf{r} = \mathbf{P}_{\text{prev}} - (\mathbf{R}\mathbf{P}_{\text{curr}} + \mathbf{t}) \tag{25}$$

Propagating uncertainty from both observations:

$$_{\mathbf{r}} = {}_{3D,\text{prev}} + \mathbf{R}\ {}_{3D,\text{curr}} \mathbf{R}^T \tag{26}$$

4. Cholesky Information Factorization

$$\mathbf{L}\mathbf{L}^T = {}_{\mathbf{r}} + 10^{-7}\mathbf{I}, \quad {}_{3D} = {}_{\mathbf{r}}^{-1} = \mathbf{L}^{-T}\mathbf{L}^{-1} \tag{27}$$

Cholesky factorization guarantees numerical stability and positive-definiteness ($< 0.1\mu$s per feature on CPU).

Pose Graph Optimization: Problem Formulation & Methods in Literature

1. The Pose Graph Optimization Problem

Given relative transformations $\mathbf{T}_{ij} \in SE(3)$ and loop closures with information weights $_{ij}$, minimize global trajectory inconsistency:

$$\min_{\{\mathbf{T}_i\}} \sum_{(i,j)\in\mathcal{E}} \left\| \log_{SE(3)} \left(\mathbf{T}_{ij}^{-1} \mathbf{T}_i^{-1} \mathbf{T}_j \right) \right\|_{ij}^2 \qquad (28)$$

where $\log_{SE(3)}(\cdot) \in \mathbb{R}^6$ maps pose errors onto the Lie algebra tangent space $\mathfrak{se}(3)$.

2. Survey of Methods Available in Literature

- **Full Bundle Adjustment (BA)** (ORB-SLAM3, COLMAP): Jointly optimizes all 6-DoF camera poses and thousands of 3D landmark positions; demands heavy multi-threaded CPU/GPU resources.
- **General Graph Optimizers** (g2o, GTSAM, Ceres): Solve large non-linear sparse normal equations via Levenberg–Marquardt; heavy memory overhead for long flights.
- **Original MAC-VO**: PyPose GPU-based Levenberg–Marquardt solver (3.8 GB VRAM).
- **Our Decoupled CPU Solver**: Decouples rotation from translation on $\mathfrak{se}(3)$, solving pose in < 2.5 ms on CPU with 0.0 MB GPU VRAM.

How We Consider IMU in Pose Graph: High-Frequency $SO(3)$ Preintegration

1. High-Rate Motion Kinematics (200 Hz)

$$\dot{\mathbf{R}} = \mathbf{R}(\boldsymbol{\omega} - \mathbf{b}_g)^{\wedge}, \quad \dot{\mathbf{v}} = \mathbf{R}(\mathbf{a} - \mathbf{b}_a) + \mathbf{g}, \quad \dot{\mathbf{p}} = \mathbf{v} \tag{29}$$

2. Discrete On-Manifold Preintegration

Between visual keyframes i and j, raw angular velocities are preintegrated directly on $SO(3)$:

$$\Delta\mathbf{R}_{ij} = \prod_{k=i}^{j-1} \mathrm{Exp}\left((\boldsymbol{\omega}_k - \mathbf{b}_{g,i})\Delta t\right) \in SO(3) \tag{30}$$

This computation is independent of the global frame, eliminating the need to re-integrate raw IMU measurements during optimization iterations.

3. First-Order Gyro Bias Correction

When the gyro bias estimate $\mathbf{b}_g$ changes by $\delta\mathbf{b}_g$ during optimization:

$$\Delta\mathbf{R}_{ij}(\mathbf{b}_{g,i} + \delta\mathbf{b}_g) \approx \Delta\mathbf{R}_{ij}(\mathbf{b}_{g,i})\,\mathrm{Exp}\left(\mathbf{J}^R_{\mathbf{b}_g}\delta\mathbf{b}_g\right) \tag{31}$$

where $\mathbf{J}^R_{\mathbf{b}_g} = \frac{\partial \Delta\mathbf{R}_{ij}}{\partial \mathbf{b}_g}$ is the analytical bias Jacobian.

4. IMU Residual in Pose Graph

$$\mathbf{r}_{\Delta R} = \log_{SO(3)}\left(\Delta\mathbf{R}_{ij}^T \mathbf{R}_i^T \mathbf{R}_j\right) \in \mathbb{R}^3 \tag{32}$$

Weighted by preintegrated covariance $\overset{-1}{\Delta R}$, constraining high-frequency rotational changes.

How We Consider IMU in Pose Graph: Sliding-Window Factor Graph

1. Sliding-Window Factor Graph

We maintain a sliding window of active keyframe poses $\mathcal{W} = \{k-M, \ldots, k\}$.

$$\min_{\mathcal{X}} \left\{ \|\mathbf{r}_{\text{prior}}\|^2 + \sum_{k\in\mathcal{W}} \|\mathbf{r}_{\text{vis},k}\|^2_{3D,k} + \sum_{k\in\mathcal{W}} \|\mathbf{r}_{\text{IMU},k}\|^2_{\text{IMU}^{-1}} \right\} \tag{33}$$

- Visual factor: 3D metric correspondence error weighted by OptiTrack-calibrated $_{3D}$.
- IMU factor: $SO(3)$ preintegrated rotation and translation kinematics.

2. Exact Schur Marginalization

When keyframe $k-M$ leaves the active window, its state $\mathbf{x}_\mu$ is marginalized without discarding information:

$$\begin{bmatrix} \mathbf{H}_{\alpha\alpha} & \mathbf{H}_{\alpha\mu} \\ \mathbf{H}_{\mu\alpha} & \mathbf{H}_{\mu\mu} \end{bmatrix} \begin{bmatrix} \delta\mathbf{x}_\alpha \\ \delta\mathbf{x}_\mu \end{bmatrix} = \begin{bmatrix} \mathbf{b}_\alpha \\ \mathbf{b}_\mu \end{bmatrix} \tag{34}$$

The Schur complement forms the new prior on remaining active states $\mathbf{x}_\alpha$:

$$\mathbf{H}_{\text{marg}} = \mathbf{H}_{\alpha\alpha} - \mathbf{H}_{\alpha\mu}\mathbf{H}_{\mu\mu}^{-1}\mathbf{H}_{\mu\alpha} \tag{35}$$

Guarantees fixed $O(1)$ CPU computation per frame.

Decoupled 6-DoF Pose Estimation Solver

Stage 1: Fast Rotation Fusion

Represent inter-frame rotations as 3D angular rotation vectors $\boldsymbol{\omega} \in \mathbb{R}^3$ (axis $\times$ angle):

- $\boldsymbol{\omega}_{\text{vis}}$: Visual rotation vector from 3D feature matching.
- $\boldsymbol{\omega}_{\text{imu}} = \int_{t_1}^{t_2} \boldsymbol{\omega}_{\text{gyro}}\, dt$: Integrated 200 Hz gyroscope.

Optimal weighted fusion of rotation vectors:

$$\boldsymbol{\omega}^* = 0.75\,\boldsymbol{\omega}_{\text{vis}} + 0.25\,\boldsymbol{\omega}_{\text{imu}} \tag{36}$$

Convert fused vector to 3×3 rotation matrix:

$$\mathbf{R}^* = \text{Rodrigues}(\boldsymbol{\omega}^*) \tag{37}$$

Physical Meaning

- Vision (75%): Provides drift-free geometric reference.
- Gyro (25%): Preserves fast angular motion dynamics.
- Rodrigues formula produces an exact orthogonal matrix.

Stage 2: Closed-Form Translation Average

With rotation $\mathbf{R}^*$ fixed, each 3D point i gives a target displacement $\mathbf{d}_i = \mathbf{P}_{\text{prev},i} - \mathbf{R}^*\mathbf{P}_{\text{curr},i}$:

$$\min_{\mathbf{t}} \sum_{i=1}^{N} w_i (\mathbf{d}_i - \mathbf{t})^T {}_i(\mathbf{d}_i - \mathbf{t}) \tag{38}$$

Exact Closed-Form Solution (No Iterations!)

$$\boxed{\mathbf{t}^* = \left(\sum_{i=1}^{N} w_i\, {}_i \right)^{-1} \left(\sum_{i=1}^{N} w_i\, {}_i \mathbf{d}_i \right)} \tag{39}$$

Why This is Ultra-Fast

- It is a standard **weighted average** in 3D!
- w_i: Robust Huber weights (suppresses outliers).
- ${}_i$: Metric information weight from Slide 12.
- Solves a single 3×3 linear system in < 0.5 ms on CPU with 0.0 MB GPU VRAM.

Experimental Setup: RealSense Sensor Rig & Live Trajectory



Figure 1: Hardware experimental rig. Intel RealSense D435if active stereo camera mounted with 4 retro-reflective OptiTrack markers.

Hardware Experimental Setup

- **Sensor**: Intel RealSense D435if (RGB-D + 200 Hz IMU).
- **Ground Truth**: 12-camera OptiTrack Prime mocap system streaming 6-DoF poses at 120 Hz (< 0.3 mm error).
- **Compute**: Single CPU thread (0.0 MB GPU VRAM).

Multi-Lap Office Walking Trajectory

- Continuous multi-lap walk: **135.56 seconds**, **3,046 frames**, **34.48 m path length**.
- Tested with active IR emitter ON to verify speckle robustness on textureless walls.

Standard IEEE RA-L Core Trajectory Metrics: ATE RMSE & RPE

1. Absolute Trajectory Error (ATE RMSE)

$$\text{ATE RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\mathbf{p}_{\text{est},i} - \mathbf{p}_{\text{gt},i}\|^2} \quad (40)$$

- **Mathematical Definition**: Global root-mean-square spatial deviation between estimated camera coordinates and ground truth across N poses after rigid $SE(3)$ Umeyama alignment.
- **Physical Meaning**: The spatial radius of error around the true trajectory over the entire mission duration.
- **Robotic Importance**: Governs global map consistency, return-to-home accuracy, and obstacle avoidance. An ATE > 0.5 m indoors causes UAVs to collide with obstacles or miss landing docks.

2. Relative Pose Error (RPE)

$$\mathbf{E}_i = (\mathbf{T}_i^{-1}\mathbf{T}_{i+\Delta})^{-1}(\mathbf{T}_{\text{gt},i}^{-1}\mathbf{T}_{\text{gt},i+\Delta}) \in SE(3) \quad (41)$$

- **Mathematical Definition**: Local transformation error over a fixed displacement/time step Δ (m/s or deg/s), evaluated without global coordinate alignment.
- **Physical Meaning**: Measures local velocity jitter and instantaneous drift rate per meter traveled.
- **Robotic Importance**: Directly drives feedback controllers (flight controller / MPC). High RPE translation/rotation jitter causes violent thrust oscillations and UAV flight destabilization.

Odometric Consistency: Path Length, Axis Drift & NEES

1. Path Length Arc Ratio ($\mathcal{L}_{\text{ratio}} = \mathcal{L}_{\text{est}}/\mathcal{L}_{\text{gt}}$)

$$\mathcal{L} = \sum_{i=1}^{N-1} \|\mathbf{p}_{i+1} - \mathbf{p}_i\|, \quad \mathbb{E}[\|\Delta\mathbf{p} + \boldsymbol{\epsilon}\|] > \|\Delta\mathbf{p}\| \tag{42}$$

- **Physical Meaning**: Ratio of total integrated trajectory arc length to true physical ground-truth path traveled.
- **Robotic Importance**: Optical flow noise has positive expected norm. Inflated path (64.8 m vs 35.0 m) indicates random-walk jitter during stops; eliminated via ZUPT.

2. Statistical Consistency (NEES)

$$\text{NEES} = (\mathbf{e}_k - \bar{\mathbf{e}})^T \, {}_k^{-1} (\mathbf{e}_k - \bar{\mathbf{e}}), \quad \mathbb{E}[\text{NEES}] \approx 2.0 \tag{43}$$

- **Physical Meaning**: Verifies whether the covariance model ${}_k$ statistically matches true error dispersion without overconfidence.

3. Zero-Velocity Updates (ZUPT) Gating

$$\Delta p_{\text{flow}} < 0.35 \text{ px} \wedge \|\boldsymbol{\omega}_{\text{corr}}\| < 0.025 \text{ rad/s} \implies \mathbf{t}_{\text{rel}} \tag{44}$$

- Halts frame-to-frame random-walk integration during stops.
- Keeps estimated path length at **34.98** m (GT: 34.97 m, 0.03% error).

4. Keyframe-Based Dynamic Baseline

- Insert keyframes dynamically when disparity > 3.5 px or rotation $> 2.5^\circ$.
- Triangulation baseline jumps from 7.6 mm $\rightarrow$ 35 mm, boosting translation SNR by $5\times$.

Metrics Explanation: Axis-Specific Drift & Hand-Eye Calibration

1. Axis-Specific Drift ($\Delta X, \Delta Y, \Delta Z$)

$$\mathbf{e}_{\text{axis}} = \frac{1}{N}\sum_{i=1}^{N} |\mathbf{p}_{\text{est},i} - \mathbf{p}_{\text{gt},i}| \in \mathbb{R}^3 \qquad (45)$$

- **Physical Meaning**: Decomposes error along camera axes (X: lateral right, Y: vertical down, Z: forward depth).
- **The 3.28 m Downward Drift Discovery**: In our multi-lap walk, net X drift was only 0.028 m and Z was 0.024 m, but Y drifted by $+3.28$ m!
- **Physical Root Cause**: Uncalibrated RealSense pitch gyro bias ($\omega_x \approx -0.25^\circ/\text{s}$) tilted orientation downward by 35° over 135 s, projecting forward motion into phantom vertical falling.
- **Fix**: 2-second stationary startup calibration $\mathbf{b}_g = \frac{1}{M}\sum \boldsymbol{\omega}_k$ eliminates Y-drift (3.28 m $\rightarrow$ 0.0099 m).

2. Hand-Eye Rigid Lever Arm Offset

$$\mathbf{p}_{\text{cam}}(t) = \mathbf{p}_{\text{marker}}(t) + \mathbf{R}_{\text{body}}(t) \cdot \mathbf{t}_{\text{hand-eye}} \qquad (46)$$

- OptiTrack tracks marker centroid; VIO tracks camera focal center.
- Solving Park-Martin $AX = XB$ eliminates physical ≈ 12 cm rotational lever-arm error during body turns.

3. Loop Closure End-to-End Drift

$$\text{Drift}_{\text{loop}} = \|\mathbf{p}_{\text{end}} - \mathbf{p}_{\text{start}}\| \qquad (47)$$

- Measures accumulated error upon returning to physical start origin.
- Critical for autonomous drone return-to-home (RTH) landing.

Experimental Validation: OptiTrack Ground-Truth Benchmark

Quantitative Metric Benchmark (135s Multi-Lap)

Metric	Baseline VIO	Enhanced VIO
ATE RMSE	1.6147 m	**0.0812** m
RPE Trans RMSE	0.3338 m	**0.0044** m
Total Path Length	64.78 m	**34.98** m

*Ground Truth Path Length: **34.97** m (OptiTrack Mocap)

Key Empirical Observations & Robotic Significance

- **Sub-Decimeter Accuracy Milestone (**8.1 **cm ATE RMSE)**: Decoupled Lie-algebra optimization and OptiTrack-calibrated metric covariance reduce global trajectory drift by **95.0**% over a continuous 135-second multi-lap flight (3,046 frames).
- **Complete Elimination of Path Inflation via ZUPT**: Baseline visual odometry accumulated **29.8** m of phantom random-walk drift during stationary intervals (64.8 m vs 35.0 m). Adaptive ZUPT gating locks estimated path length to **34.98** m (0.03% error).
- **Control-Ready Velocity Smoothness (**4.4 **mm RPE)**: Relative translation jitter dropped by **98.7**%, eliminating the high-frequency velocity noise that causes thruster oscillations and flight instability in PX4/ArduPilot autopilots.
- **Edge Hardware Viability (**0.0 **MB GPU VRAM)**: Runs at **32.4** FPS with latency < 2.5 ms on CPU, leaving edge companion GPUs (Jetson / RPi 5) 100% free for real-time dense mapping and obstacle avoidance.

Future Work: Algorithmic Refinement & Drone Deployment

1. Improving State Estimation Accuracy

- **Online Accelerometer Bias Estimation**: Integrate continuous accelerometer scale and bias estimation into the sliding window to improve metric scale under violent acceleration.
- **Multi-Camera Visual-Inertial Fusion**: Extend formulation to dual-camera rigs (forward RGB-D + downward optical flow camera) to guarantee observability during aggressive drone pitching.

2. Deployment on Edge Computers

- Target platforms: **NVIDIA Jetson Orin Nano** and **Raspberry Pi 5**.
- Exploit 100% CPU execution (0.0 MB GPU VRAM) to leave the onboard NVDLA/GPU free for real-time dense 3D mapping and YOLO obstacle detection.

3. Publishing Estimated Position to Drone Autopilot

- **Autopilot Integration**: Stream real-time 6-DoF state estimate directly to **PX4 / ArduPilot** flight controllers via MAVLink (`VISION_POSITION_ESTIMATE`).
- **ROS2 Interface**: Publish to `/mavros/vision_pose/pose` at 30–40 Hz with latency < 25 ms.
- **Autonomous Capabilities**:
  - GPS-denied autonomous indoor hover and station-keeping.
  - Multi-waypoint 3D flight in unknown indoor warehouses.
  - Precision autonomous landing on docking pads without external positioning.

Key Scientific Literature & Foundational References

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3. **ORB-SLAM3**: Campos, C., et al., "ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual–Inertial, and Multimap SLAM," *IEEE Transactions on Robotics (T-RO)*, vol. 37, no. 6, pp. 1874–1890, 2021.
4. **VINS-Mono**: Qin, T., Li, P., and Shen, S., "VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator," *IEEE Transactions on Robotics (T-RO)*, vol. 34, no. 4, pp. 1004–1020, 2018.
5. **Lucas–Kanade**: Lucas, B. D., and Kanade, T., "An Iterative Image Registration Technique with an Application to Stereo Vision," *IJCAI*, pp. 674–679, 1981.
6. **Umeyama Alignment**: Umeyama, S., "Least-Squares Estimation of Transformation Parameters Between Two Point Patterns," *IEEE Trans. Pattern Anal. Mach. Intell. (PAMI)*, vol. 13, no. 4, pp. 376–380, 1991.
7. **Hand-Eye Calibration**: Park, F. C., and Martin, B. J., "Robot Sensor Calibration: Solving AX= XB on the Euclidean Group," *IEEE Transactions on Robotics and Automation (T-RA)*, vol. 10, no. 5, pp. 717–721, 1994.

Thank You!

MAC-RGBD-VIO: Real-Time Visual-Inertial Odometry via Decoupled Lie-Algebra Optimization & Analytical Metric Covariance

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Summary of Core Deliverables

- **0.0 MB GPU VRAM**: 100% CPU execution at > 30 FPS.
- **OptiTrack Calibrated**: Statistically grounded covariance (NEES ≈ 1.87).
- **95.0% Error Drop**: ATE reduced from 1.61 m $\rightarrow$ **0.081** m.
- **Active IR Compatible**: Stable on plain featureless walls.

Research & Deployment Contacts

- **Laboratory**: Robotics Research Laboratory
- **Target Journal**: IEEE Robotics and Automation Letters (RA-L)
- **Source Code**: Modular Python / C++ on GitHub
- **Hardware**: Intel RealSense D435if + PX4 Autopilot

Questions & Discussion